

Contextualized Anomaly Analysis in Country-Risk Time Series Using Historical Precedents

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Abstract—Anomaly analysis in financial time series is challenging when extreme variations are driven by political, economic, and social events rather than by signal dynamics alone. This work-in-progress presents a framework for contextualized anomaly analysis of Ecuador’s daily country-risk series. The proposal combines consensus-based historical anomaly detection, exploratory generation of synthetic short-term scenarios, retrieval of historical precedents through numerical and visual similarity, and semantic enrichment using an OWL/RDF knowledge graph linked to categorized news. A preliminary run over data from 2004 to 2026 produced 162 consolidated historical anomalies and 149 synthetic anomaly candidates. Initial results suggest that ranked historical precedents can support expert interpretation by connecting anomalous patterns with prior events and contextual evidence. Further work is required on threshold calibration, forecasting validation, and reproducible news collection.

Index Terms—Time series anomaly detection, country risk, contextualized analytics, dynamic time warping, semantic enrichment, knowledge graphs.

I. INTRODUCTION

Sovereign risk serves as a composite indicator of macroeconomic stability, political volatility, and international market perceptions. In economies exposed to exogenous factors, such as Ecuador, the daily country risk series exhibits abrupt episodes that cannot be explained solely by the internal dynamics of the signal. Although the detection of anomalies in time series has advanced through statistical, density-based, and deep learning approaches [2], [6], the numerical identification of an atypical event is insufficient for expert analysis if it is not linked to historical precedents and contextual evidence.

The detection of similar patterns can rely on temporal alignment methods such as Dynamic Time Warping (DTW) [1], while visual models such as CLIP allow for the representation of anomaly graphs using comparable embeddings [4]. However, these approaches are typically used separately and rarely integrate with external context and semantic traceability. OWL/RDF ontologies offer an alternative for representing series, anomalies, methods, news, and similarity results as queryable entities in a knowledge graph [8], [20].

In this work, country risk is treated as a financial-market indicator that reflects sovereign-risk perception and external financing conditions. Its variations may be associated with macroeconomic, political, security, and international events; however, these factors are not modeled as explanatory variables in this preliminary version. Instead, they are used as contextual dimensions for interpreting anomalous episodes through

categorized news and historical precedents. The proposal integrates: i) retrospective detection based on consensus among eight heterogeneous methods; ii) exploratory generation of short-term synthetic scenarios; iii) retrieval of historical precedents using numerical and visual similarity; and iv) semantic enrichment with news categorized in an OWL/RDF database. The objective is not to validate crisis predictions, but to transform observed or potential anomalies into traceable queries of contextualized historical precedents.

II. BACKGROUND AND RELATED WORKS

A *time series* is a chronologically ordered sequence of observations, and an *anomaly* corresponds to a point or segment whose behavior deviates from the expected pattern of the series [14], [15]. In time series, anomaly detection seeks to identify these atypical episodes and is typically addressed using three families of methods: distance-based approaches, which compare subsequences based on their morphological similarity; density-based approaches, which identify isolated regions in the representation space; and prediction- or reconstruction-based approaches, which use model error as a signal of deviation [11]. This methodological diversity motivates the use of consensus schemes, especially in noisy financial domains where a single detector can induce false positives.

Time series forecasting estimates future values based on historical observations and can be performed using statistical, machine learning, or deep learning models [16], [21]. In this work, these models are not used as a definitive mechanism for crisis prediction, but rather as exploratory generators of short-term synthetic trajectories. These trajectories allow for similarity queries against a corpus of previously consolidated historical anomalies.

Finally, OWL/RDF ontologies allow for the formal representation of entities, properties, and relationships within a domain in machine-readable semantic graphs [20]. RDF models knowledge using subject–predicate–object triples, while SPARQL enables the querying and integration of this information. This representation is suitable for linking series, anomalies, detection methods, synthetic scenarios, news, and similarity results in a traceable knowledge base.

The literature on anomalies in univariate time series has focused primarily on detection, while event comparison, structured storage, and external contextualization are often addressed only partially or independently [9], [14], [19]. Recent reviews identify interpretability and traceability as open

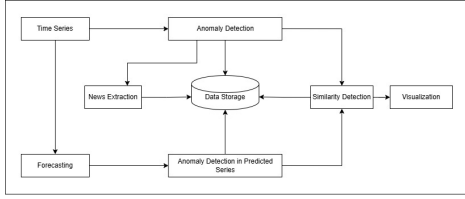


Figure 1. General architecture for context-aware detection, generation of synthetic scenarios, and retrieval of historical precedents.

challenges in the field [11], and warn that isolated detection is insufficient when contextual explanation mechanisms are not incorporated [10]. Among integrative approaches, Sintel combines detection, visualization, and structured signal storage, although it does not address precedent retrieval or contextualization via external sources [18]. Calikus et al. propose automatic context discovery for anomalies, but without integrating it with semantic representation or synthetic scenarios [12]. For their part, Park et al. explore transformations from time series to images to improve detection, suggesting the potential of visual representations for comparing morphological patterns [13]. Taken together, these works demonstrate significant progress, but leave open the possibility of simultaneously integrating historical detection, exploratory scenario generation, precedent retrieval, external context, and semantic representation within a single analytical framework.

III. FRAMEWORK FOR GENERATING SYNTHETIC SCENARIOS AND RETRIEVING HISTORICAL DATA

Fig. 1 summarizes the proposed architecture. The framework takes the historical country risk series as input and organizes the analysis into two complementary workflows. The retrospective workflow builds a corpus of historical anomalies through consensus-based detection, enrichment with news, and semantic representation in an OWL/RDF database. The exploratory flow generates short-term synthetic trajectories, detects candidate anomalies, and uses them as queries to retrieve comparable historical precedents. The comparison is based on numerical similarity, visual similarity, and thematic context, with the aim of producing a traceable ranking of analogous events rather than a binary decision.

The main components are described below:

A. Study Time Series

The dataset was selected based on three criteria: historical availability, temporal granularity, and the ability to contextualize the data using external sources. After comparing national oil production, intentional homicides, and country risk, the latter was selected for its daily coverage since 2004, its expression in basis points (bps), and its sensitivity to macroeconomic, political, and financial events documented in external sources. The series, published by the Central Bank of Ecuador, serves as the common input for the retrospective detection of anomalies, the exploratory generation of synthetic scenarios, and the semantic enrichment of the identified events.

B. Anomaly Detection in the Original Series

Retrospective detection is applied to the original time series using eight detectors running independently: ARIMA, DIF, COUTA, DBSCAN, Isolation Forest, TranAD, TimesNet, and TimeGPT. This selection combines statistical, density-based, machine learning, and deep learning approaches to reduce reliance on a single detection criterion. Each method produces point-level anomaly candidates, which are subsequently grouped into consecutive time intervals defined by a start date and an end date.

Segments obtained by different methods are merged when they overlap in time or share a start date, while retaining the set of detectors that participated in each event. A consolidated segment is accepted as a historical anomaly if the number of methods reporting it satisfies the threshold $c_{\min} = \text{CONSENSO_MIN_ALGOS}$. In this preliminary version, $c_{\min} = 2$ was set, a value selected exploratorily as a compromise between coverage and reliability: a higher threshold would excessively reduce the corpus available for comparison, while $c_{\min} = 1$ would be equivalent to accepting individual detections without cross-validation. Each accepted anomaly is associated with a temporal context window, which is used in the subsequent enrichment and comparison stages.

In the preliminary run, the eight detectors generated 1,745 raw anomaly points; after clustering and consensus-based consolidation, 162 historical anomalies were identified for the period 2005–2025.

C. News Extraction and Categorization

News collection followed a predefined manual protocol. For each anomaly, searches were conducted within the interval $[t_i - \delta, t_f + \delta]$, with $\delta = 5$ days, using combinations of terms related to Ecuador, country risk, debt, oil, government, security, and economic or political crises. Items were retained when they fell within the temporal window, referred to Ecuador or to external events directly relevant to Ecuadorian country risk, and matched at least one predefined thematic category. Duplicates, opinion pieces without factual context, and unrelated items were excluded. Each retained item was stored with date, title, source, link, summary, and category.

The predefined categories include economy and oil, politics and government, security and crime, international relations, energy issues, environment and disasters, drug trafficking, and country risk or quality of life.

D. Storage of Anomalies and Related News

The knowledge base was implemented using an OWL/RDF ontology serialized in Turtle, hosted on Apache Jena Fuseki. It contains twelve classes: Series, Point, Anomaly, Source, NewsArticle, Category, DetectionMethod, PredictionMethod, Similarity, SimilarityMethod, SimilarityRun, and SelectionPolicy. The ontology was populated using the `semantic_exporter.py` module, which generates eight separate TTL files: original series, series points, detected anomalies, synthetic series, synthetic points, synthetic anomalies, visual similarity

results, and graphical similarity results, all of which are loaded into Fuseki for querying via SPARQL.

The CLIP model (ViT-B/32) is used to generate vector representations of the PNG graphics for each anomaly. The images are generated with a fixed Y-scale based on the overall range of the series, ensuring that the scale is comparable across anomalies and preventing nearly flat series from being visually confused with actual drops.

E. Generation of synthetic scenarios

Based on the historical time series, short-term synthetic trajectories are generated that serve as exploratory queries against the corpus of established historical anomalies. The objective of this module is not to demonstrate crisis prediction, but rather to produce candidate patterns that can be compared with observed precedents using numerical, visual, and contextual similarity.

Seven forecasting models are used: ARIMA, SVR, Random Forest, Prophet, LSTM, N-BEATS, and DeepAR. Each model is run in two modes: *DIRECT*, where the entire horizon is predicted in a single operation, and *FEEDBACK*, where each estimated value is incorporated as input to predict the next step. This yields fourteen synthetic trajectories, one for each model-mode combination, with a 30-day horizon. The generated values are rounded before storage to maintain the discrete scale of the original series and reduce spurious detections due to marginal numerical variations.

F. Detection of Candidate Anomalies in Synthetic Scenarios

The same set of eight detectors used in the original series is applied to each synthetic trajectory. However, since these values correspond to generated scenarios rather than actual observations, the detections are interpreted as candidate anomalies. Unlike the retrospective workflow, no consensus-based consolidation is applied across methods; each detector groups its own anomalous points into consecutive segments, maintaining the association with the predictive model and the generation modality. This decision preserves the diversity of patterns produced by each model-detector combination and allows them to be used later as individual queries against the corpus of consolidated historical anomalies.

G. Retrieving Precedents Based on Similarity

The retrieval of precedents is implemented in two scenarios. The first compares candidate anomalies detected in synthetic trajectories against the corpus of consolidated historical anomalies. The second operates exclusively on observed data, separating recent and older anomalies from the original series to assess whether the most recent events find plausible analogies in the past. In both cases, the result is not interpreted as a binary classification, but rather as a ranking of precedents supported by numerical, visual, and thematic evidence.

1) *Numerical Similarity*: For each anomaly, a time vector is constructed from the anomalous segment or from a fixed-size context window, normalized prior to comparison. The numerical similarity combines two components: a morphological component based on DTW, which allows for the comparison

of patterns with local temporal shifts, and a structural similarity based on descriptive features of the event, such as duration, minimum value, maximum value, amplitude, mean, standard deviation, energy, and slope. The final score is defined as:

$$s_{num} = \alpha s_{DTW} + (1 - \alpha) s_{feat},$$

where s_{DTW} represents shape similarity, s_{feat} represents feature vector similarity, and α controls the relative weight of temporal morphology. Only pairs that exceed a configurable minimum threshold are retained, and the candidates are sorted from highest to lowest similarity.

2) *Visual Similarity*: Visual similarity uses embeddings generated from normalized graphs of each anomaly. For each pair of anomalies, the cosine similarity between their visual vectors is calculated. Given a query event, the *top-K* closest precedents from the historical corpus are retrieved, and ambiguous matches are discarded when the difference between the first and second candidates is less than a predefined margin. This step does not replace numerical comparison but serves as complementary evidence to identify morphologically similar patterns from a comparable visual representation.

H. Visualization and Semantic Querying

The pipeline's results are integrated into an interactive dashboard designed for the inspection and traceability of anomalies. The interface allows users to explore the historical time series, synthetic trajectories, detected anomalous segments, and retrieved precedents based on numerical and visual similarity. In addition, it incorporates a semantic query view connected to Apache Jena Fuseki, from which detection methods, temporal distributions of anomalies, similarity rankings, associated news, and thematic categories can be examined without requiring SPARQL queries from the user. This layer does not modify the analytical results, but rather facilitates their qualitative review and the verification of the evidence used to contextualize each precedent.

IV. PRELIMINARY RESULTS

Table I summarizes the initial results of the pipeline. The country risk series contains 7,854 daily records for the period 2004–2026. Based on the observed series, the eight detectors generated 1,745 raw anomaly points; after temporal grouping and consensus-based consolidation, 162 historical anomalies were identified for the period 2005–2025. This corpus constitutes the retrospective basis for the stages of contextualization, semantic storage, and precedent retrieval.

In the exploratory phase, 14 synthetic 30-day trajectories were generated, corresponding to seven forecasting models run in two modes. A total of 149 candidate anomalies were identified in these trajectories. When compared with the 162 consolidated historical anomalies, numerical similarity yielded 23,976 pairs with $s_{num} \geq 0.50$, with an average of 0.855 and a maximum value of 0.966. Since this threshold retrieves a very high proportion of the possible pairs, this result is interpreted as an initial high-coverage configuration and not as definitive evidence of discrimination. The next phase of

Table I
PRELIMINARY RESULTS OF THE PIPELINE

Component	Result
Daily records in the series	7,854
Covered period	2004–2026
Detection methods	8
Raw outliers	1,745
Consolidated historical anomalies	162
Synthetic trajectories generated	14
Forecast horizon	30 days
Synthetic candidate anomalies	149
Numerical pairs with $s_{num} \geq 0.50$	23,976
Synthetic anomalies with valid visual pairs	116/149
Valid visual pairs	228

the work requires calibrating more selective thresholds and evaluating the quality of the ranking through expert validation or top- K metrics.

Visual comparison using embeddings retrieved up to two precedents for each synthetic anomaly. Of the 149 candidate anomalies evaluated, 116 obtained at least one valid visual precedent, generating 228 pairs with an average similarity of 0.947 and a maximum of 0.969. These results suggest that visual representation can provide complementary evidence for prioritizing morphologically similar patterns, although its use should be considered supplementary to numerical comparison and contextual review.

As an exploratory example, a one-off anomaly detected by DIF on the *PROPHET_DIRECT* for the date 2026-02-10 retrieved as close precedents episodes from December 2019 and February 2016, both with numerical similarity exceeding 93% and visual similarity close to 96%. This result should not be interpreted as predictive validation of a future event, but rather as evidence that the framework can transform a candidate anomaly into a query for plausible historical precedents.

Additionally, a purely retrospective scenario was evaluated, separating recent and older anomalies within the observed series. In this test, recent events from 2025 retrieved precedents from 2023 with high visual similarity and thematic matches in economics, politics, security, and country risk. This behavior suggests that the retrieval engine can identify not only comparable temporal patterns but also contextual recurrences associated with the events.

Finally, the OWL/RDF layer enabled the structured storage of detection results, similarity scores, and news context. This facilitates traceable queries regarding which historical anomalies resemble a given episode, which methods were involved in its detection, and which news categories were associated with it. From an analytical perspective, this traceability constitutes an advantage over pipelines that deliver only anomaly or similarity scores without organized contextual evidence.

V. DISCUSSION, CONCLUSIONS, AND FUTURE WORK

The proposed framework should be understood as an exploratory decision-support tool for contextual anomaly analysis, not as a validated crisis-prediction model. Its main contribution is the transformation of anomalous movements in a country-risk time series into traceable queries over historical precedents, similarity scores, and contextual news

evidence. The current thresholds, including the consensus value, news window, and similarity cutoffs, should be interpreted as exploratory parameters rather than optimized values. The large number of retrieved numerical pairs suggests that the present configuration favors coverage over discrimination. Future work will therefore focus on temporal backtesting with sliding windows, stricter threshold calibration, Top-K ranking metrics, baseline comparisons, and expert qualitative review of the retrieved precedents and their contextual evidence.

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